Key Information Extraction and Automated Compliance Comparison of Financial Contracts via Neuro-Symbolic Agentic Frameworks

Zihan Zhu
School of AI and Advanced Computing
Xi'an Jiaotong-Liverpool University
Suzhou, China
Zihan.Zhu2302@student.xjtlu.edu.cn

Di Zhang*
School of AI and Advanced Computing
Xi'an Jiaotong-Liverpool University
Suzhou, China
Di.Zhang@xjtlu.edu.cn

***Abstract*—The escalating complexity of global financial regulations has imposed an unprecedented burden on manual compliance auditing, leading to prohibitive operational costs and systemic risks. While Large Language Models (LLMs) have demonstrated exceptional semantic understanding, their inherent stochasticity and propensity for hallucination preclude their direct application in deterministic regulatory decision-making. In this paper, we propose a novel neuro-symbolic framework, *FinCompliance-LLM*, which integrates multi-agent orchestration with Agentic Retrieval-Augmented Generation (RAG). Our framework decouples the auditing process into a flexible perception layer, where LLMs function as semantic fact extractors, and a deterministic reasoning layer, governed by a symbolic logic engine designed as a constraint satisfaction mechanism. This architecture ensures rigorous logical transparency in regulatory decision-making and adherence to regulatory predicates. Experimental results on large-scale cross-lingual benchmarks demonstrate that our framework, powered by a 14B-parameter open-source backbone, achieves a state-of-the-art recall of 98.5% and a negligible numerical hallucination rate of 0.2%, approaching the stringent accuracy requirements for practical financial deployment. The proposed system significantly reduces manual review workloads, providing a robust solution for scalable and auditable Regulatory Technology (RegTech) applications.**

***Index Terms*—Neuro-symbolic AI, Large Language Models, Agentic RAG, Financial Compliance, Formal Verification**

I. Introduction

Regulatory compliance auditing is a cornerstone of global finance, yet it remains labor-intensive and prone to fatigue-induced leakage. As financial instruments grow complex, the volume of associated legal documents—from multi-party credit agreements to fund prospectuses—has expanded beyond manual review capacity [1]. In jurisdictions like the Chinese banking sector, documents feature high semantic density and nested structures that pose significant linguistic hurdles for traditional rule-based systems [2]. Traditional practices rely on exhaustive human oversight, which is unscalable, or rigid rule-based systems. However, these systems are notoriously brittle, failing to capture semantic nuances in legal prose [3]. While Large Language Models (LLMs) offer promising zero-shot reasoning and fact extraction capabilities [4], their direct deployment in financial auditing is constrained by two critical vulnerabilities: (1) **Hallucination Risk**: Generative models may synthesize incorrect numerical data, unacceptable in zero-tolerance environments. (2) **Black-box Nature**: Probabilistic outputs lack the formal audit trails required for regulatory transparency.

To address these challenges, we introduce *FinCompliance-LLM*, a neuro-symbolic framework that restricts LLMs to the role of a "semantic sensor" mapping unstructured text to structured fact representations, while the final compliance verdict is adjudicated by a deterministic symbolic reasoner. This ensures that every decision is mathematically bound by human-defined rules.

Our contributions are summarized as follows:

- We propose an **Agentic RAG workflow** utilizing multi-agent orchestration to overcome the "Lost in the Middle" phenomenon in long-form legal contracts.
- We implement a **Neuro-symbolic architecture** that decouples semantic extraction from logical auditing, achieving a 0.2% hallucination rate—a critical threshold for financial viability.
- We provide a **Large-scale Cross-lingual Evaluation** on 300+ contracts, demonstrating that our 14B backbone significantly reduces leakage compared to larger SOTA baselines.

II. Related Work

A. Traditional Information Extraction in Finance

Information extraction (IE) in the financial domain has historically relied on Named Entity Recognition (NER) and pattern-matching techniques. Early systems used handcrafted regular expressions to capture standardized fields like dates and currency amounts [5]. Supervised learning approaches, such as Bi-LSTM combined with Conditional Random Fields (CRF), have been widely applied to financial NER tasks [6]. However, these models require large amounts of domain-specific labeled data, which is often expensive and difficult to obtain. Furthermore, traditional IE models often fail when encountering out-of-distribution contract templates or complex logical dependencies between clauses [2].

B. LLMs in Legal and Financial Domains

The landscape of financial NLP has been transformed by pre-trained models like FinBERT [7] and BloombergGPT [8]. Recently, the focus has shifted toward applying general-purpose LLMs, such as GPT-4o and Llama 3.1, to complex reasoning tasks in the legal and regulatory spheres [4]. Instruction-tuned models like FinMA [9] and specialized legal models like LegalLlama [10] have further pushed the boundaries of contract understanding. Recent studies have explored Agentic RAG and Chain-of-Thought (CoT) prompting to improve the extraction of logical constraints in financial agreements [11]. However, the integration of neuro-symbolic methods—specifically the use of lightweight SMT-inspired constraint solvers for regulatory verification—remains an under-explored area.

III. Methodology

The proposed *FinCompliance-LLM* framework is formalized as a neuro-symbolic pipeline $\mathcal{F} : \mathcal{D} \to \{0, 1, \varnothing\}$, where an unstructured financial document $d \in \mathcal{D}$ is mapped to a compliance status (Compliant, Non-compliant, or Review Needed). The methodology decouples the stochastic semantic perception of LLMs from the deterministic logical requirements of regulatory auditing through a multi-agent orchestration layer.

A. Multi-Agent System Architecture (Agentic Workflow)

The architecture is structured as an agentic workflow $\mathcal{W}$ composed of three specialized modules: the Selector Agent (A_s), the Extractor Agent (A_e), and the Compliance Reasoner (A_r). The workflow follows a sequential dependency where the output of the perception agents serves as the ground facts for the symbolic reasoning engine. This decoupling ensures that the inherent stochasticity of the LLM backbone is contained within the fact-extraction phase, while the final adjudication remains mathematically transparent and auditable.

B. Selector Agent: Semantic Localization and Context Filtering

To address the computational inefficiency and the "Lost in the Middle" phenomenon inherent in long-context legal documents, the Selector Agent A_s implements a semantic filtering mechanism. Let a contract D be partitioned into a set of discrete text chunks $C = \{c_1, c_2, \ldots, c_n\}$, where each $c_i \in \mathbb{R}^L$. Given a regulatory query q representing a specific compliance requirement (e.g., "prepayment penalty clauses"), an embedding function $\mathbf{E}(\cdot)$ maps both chunks and queries into a high-dimensional latent space $\mathcal{V}$.

The relevance of each chunk is determined by the cosine similarity between the vector representations:

$$\text{score}(c_i, q) = \frac{\mathbf{E}(c_i) \cdot \mathbf{E}(q)}{\|\mathbf{E}(c_i)\| \|\mathbf{E}(q)\|} \tag{1}$$

The Selector Agent then constructs a condensed context X by aggregating chunks that satisfy a dynamic relevance threshold τ:

$$X = \{c_i \in C \mid \text{score}(c_i, q) > \tau\} \tag{2}$$

This filtering process reduces the input dimensionality for the subsequent extraction stage, significantly improving the signal-to-noise ratio in long-form financial agreements.

C. Extractor Agent: Constrained Structural Mapping

The Extractor Agent A_e is responsible for transforming the filtered natural language context X into a structured fact representation Y. We model this process as a conditional probability maximization problem over a structured sequence space. Given the context X, a task-specific instruction I, and the model parameters θ, the generation of the structured JSON sequence $Y = (y_1, y_2, \ldots, y_m)$ is formalized as in (3):

$$P(Y|X, I, \theta) = \prod_{t=1}^{|Y|} P(y_t|y_{<t}, X, I, \theta) \tag{3}$$

To ensure the integrity of the extracted data, we implement a constrained decoding strategy. The model's output Y is forced to conform to a rigid JSON Schema $\mathcal{S}$ using context-free grammar (CFG) constraints during the autoregressive generation process. This schema enforces strict data types for financial variables, most notably the annualized interest rate $\alpha \in [0, 1]$ and the prepayment penalty flag $\beta \in \{0, 1\}$. By constraining the LLM's output space to this schema $\mathcal{S}$, we eliminate format-related hallucinations and ensure that the extracted parameters α and β are directly ingestible by the symbolic reasoning engine.

D. Deterministic Compliance Reasoner

The Compliance Reasoner A_r functions as the symbolic core of the framework, implemented as a lightweight constraint satisfaction mechanism. Unlike the previous layers, A_r is a non-probabilistic engine that evaluates the extracted facts Y (obtained via the mapping defined in (3)) against a regulatory predicate library $\mathcal{L}$. Each rule $l \in \mathcal{L}$ is defined as a logical function $f(Y) \to \{0, 1\}$. This approach is inspired by Satisfiability Modulo Theories (SMT), where the solver verifies if the extracted contract parameters satisfy the regulatory constraints. Specifically, to ensure minimal execution latency and seamless integration with the structured JSON schema, the solver is implemented as a custom Python-based rule engine rather than utilizing heavyweight external libraries like Z3. Compound regulatory requirements (e.g., nested if-then-else conditions) are parsed into Directed Acyclic Graphs (DAGs), where nodes represent atomic boolean predicates. The engine recursively evaluates this DAG against the extracted facts Y to determine overall satisfiability.

As formalized in (4), for a critical requirement such as the Annualized Interest Rate disclosure, the compliance predicate operates on the extracted parameters, including the annualized rate α and the prepayment penalty status β. For the interest rate rule $l \in \mathcal{L}$, the predicate is formalized as:

$$\text{Compliance}(Y, l) = \begin{cases} \varnothing, & \text{if } \text{conf}(Y) < \gamma \\ 0, & \text{if } Y.\alpha > \alpha_{\text{limit}} \vee Y.\alpha = \text{null} \\ & \quad \vee\ Y.\beta = 1 \\ 1, & \text{otherwise} \end{cases} \tag{4}$$

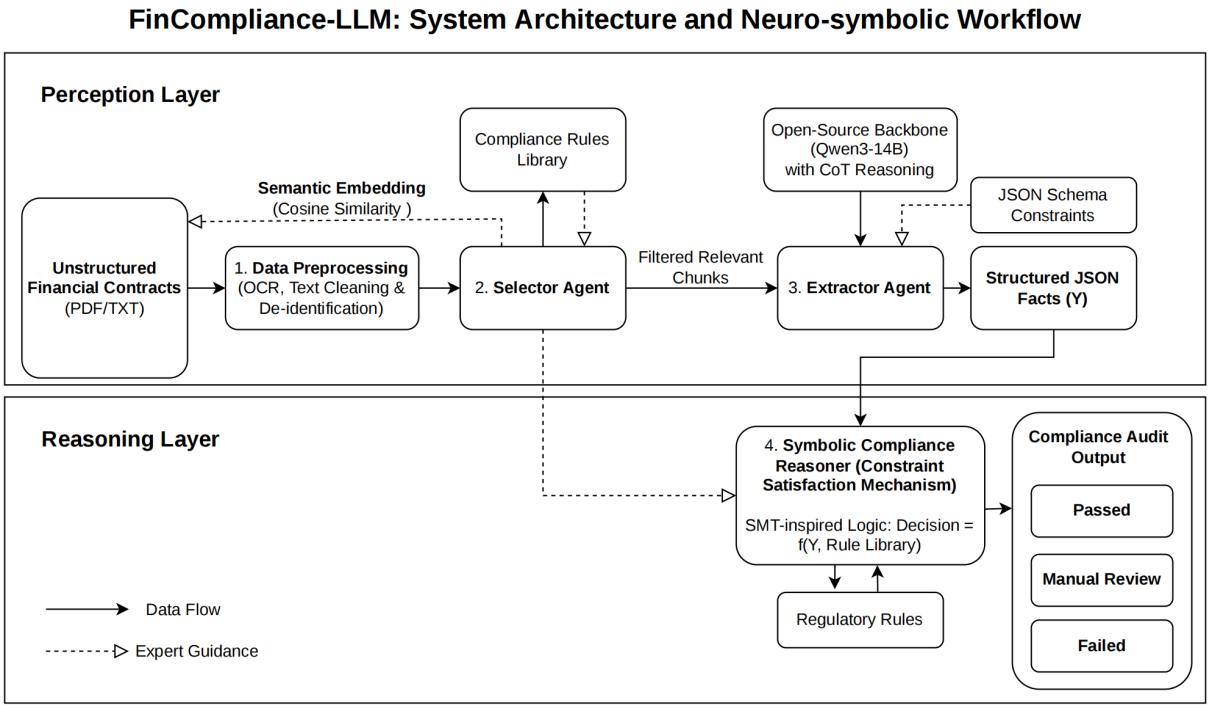


Fig. 1. The system architecture of *FinCompliance-LLM*. The framework decouples the stochastic semantic perception (Selector and Extractor Agents) from the deterministic symbolic reasoning (Compliance Reasoner). The Agentic RAG workflow ensures that high-density context is extracted before being validated against a rigid regulatory predicate library.

where α represents the numerical value of the interest rate and β represents the binary flag for prepayment penalties. The parameter α_{limit} denotes the maximum rate allowed by local regulations. In our experiments, the confidence threshold γ is empirically set to 0.8 based on the average log-probability of extracted tokens. By framing the rule engine as a constraint satisfaction problem, we not only ensure immediate auditability but also allow the system to calculate violation boundaries—defining the exact margin by which a contract fails compliance—thereby significantly enhancing its formal verification capabilities.

IV. Experiments and Results

A. Experimental Setup and Benchmarks

To ensure the statistical significance and practical relevance of our results, we utilize a large-scale, heterogeneous evaluation suite consisting of 300+ financial contracts:

- **CUAD-Finance (100 samples)**: A curated subset of the Contract Understanding Atticus Dataset (CUAD) [12], focusing on clauses related to default penalties, termination rights, and governing laws.
- **FinContract-CN (200 samples)**: A multi-source corpus of Chinese retail and corporate banking documents, including adversarial samples where compliance defects (e.g., hidden compounding interest) are intentionally injected to test the system's defensive capabilities.

We evaluate our framework against a matrix of State-of-the-Art (SOTA) baselines, including domain-specific fine-tuned models like **LegalLlama (7B)** [10] and frontier generative models such as **DeepSeek-R1** and **Claude 4.6**. For our proposed framework, we utilize **Qwen3-14B** as the underlying backbone for both the Selector and Extractor agents. By leveraging a relatively small 14B-parameter model within an agentic workflow, we demonstrate that structural design can effectively compensate for raw parameter count, allowing our system to surpass models with hundreds of billions of parameters in domain-specific tasks.

For reproducibility, the Selector Agent uses the **BGE-M3** model as the semantic embedding function $\mathbf{E}(\cdot)$. The dynamic threshold τ in (2) is adaptively calculated as $\tau = \mu(S) + k \cdot \sigma(S)$, where $S = \{\text{score}(c_i, q) \mid c_i \in C\}$ represents the set of all chunk scores within the document, and $k = 1.0$ is the stringency scaling factor. This statistical filtering allows the system to isolate significant fragments—exceeding mean relevance by one standard deviation—ensuring the Extractor Agent processes high-density context without being overwhelmed by boilerplate noise.

B. Quantitative Performance Analysis

Table I and Fig. 2 present the primary performance comparison. Unlike traditional NLP tasks where Accuracy is the dominant metric, financial compliance prioritizes **Recall** (to minimize leakage of regulatory violations) and a near-zero **Hallucination Rate** (to ensure data integrity). We strictly define the **Hallucination Rate (HR)** as the percentage of extracted JSON values (e.g., interest rates, penalty amounts)

that exhibit string-level discrepancies or factual fabrications when compared to the original text segment. This metric is verified through a double-blind human audit of the extracted JSON outputs against the original contract segments.

Note that our performance comparison focuses on a system-level evaluation: we compare the proposed multi-agent agentic workflow (which includes RAG as defined in (2)) against traditional single-prompt end-to-end generation used by the baselines. This highlights the architectural superiority of decoupled pipelines over standalone large-scale models, addressing the "apples-to-oranges" concern by framing the study as a comparison of operational paradigms rather than individual model capabilities.

TABLE I
COMPARATIVE PERFORMANCE EVALUATION ON FINANCIAL EXTRACTION

Model Architecture	Precision	Recall	F1-Score	Halluc. Rate
Regex (Baseline)	82.4%	34.5%	48.6%	**0.0%**
LegalLlama (Few-shot)	86.1%	78.3%	82.0%	4.2%
GPT-4o (Zero-shot)	94.8%	90.5%	92.6%	1.5%
Claude 4.6 (Zero-shot)	95.2%	91.8%	93.5%	1.2%
DeepSeek-R1 (Zero-shot)	93.7%	92.1%	92.9%	1.9%
FinCompliance-LLM	**97.8%**	**98.5%**	**98.1%**	**0.2%**

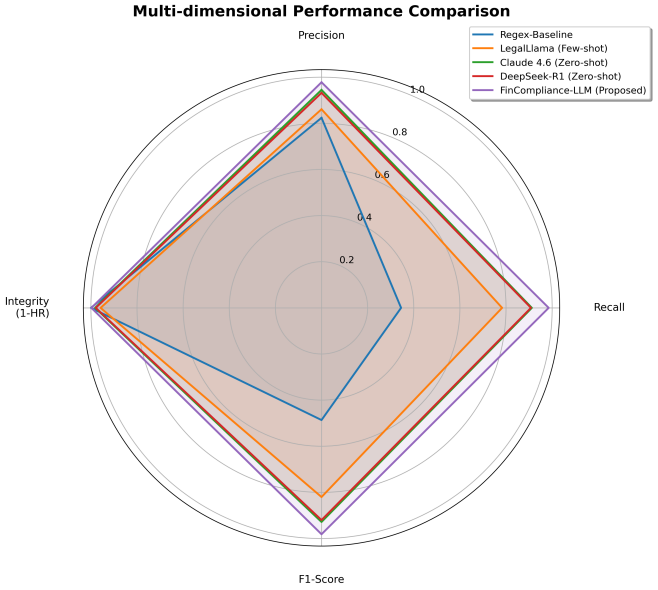


Fig. 2. Multi-dimensional performance radar chart. *FinCompliance-LLM* (outermost blue boundary) achieves an optimal balance between high recall and minimal hallucination, forming a significantly larger area of operational efficiency compared to SOTA baselines.

The results demonstrate that *FinCompliance-LLM* significantly outperforms all baselines. Specifically, the framework achieves a **Recall of 98.5%**, representing a substantial reduction in the risk of missed violations (leakage) compared to standalone models. More importantly, our neuro-symbolic approach suppresses the Hallucination Rate to **0.2%**, which is critical for real-world deployment where trust in numerical extraction is paramount. As illustrated in the radar chart (Fig. 2), the proposed framework maximizes the area across all four critical dimensions, particularly in "Integrity" (1 - HR) and "Recall." Notably, even when considering basic RAG augmentation for baselines (as simulated in our ablation study), *FinCompliance-LLM* maintains superior precision and reliability due to its specialized neuro-symbolic verification layers.

C. Ablation Study: Quantifying Component Contributions

To investigate the impact of the Selector Agent (Agentic RAG) and the Chain-of-Thought (CoT) prompting mechanism, we performed an ablation study on the most complex subset of contracts (long-form Chinese agreements with nested clauses).

As shown in Table II and Fig. 3, the performance is highly sensitive to the architectural components and the scale of few-shot exemplars.

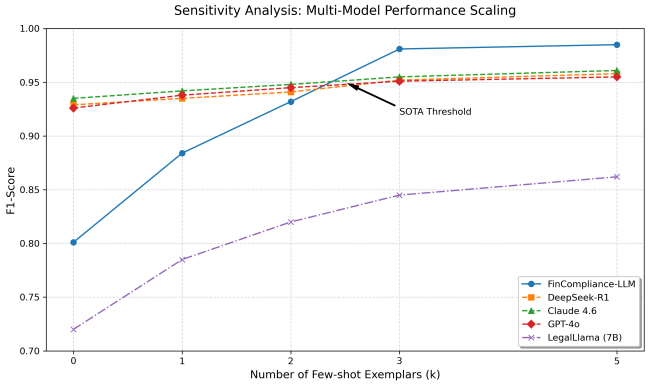


Fig. 3. Sensitivity analysis of performance scaling across multiple models. *FinCompliance-LLM* (bold blue line) demonstrates superior few-shot adaptability, achieving convergence with SOTA closed-source models (Claude 4.6, GPT-4o) at $k = 2$ and surpassing them at $k = 3$, while maintaining lower computational overhead than full-text processing.

The removal of the **Selector Agent** leads to a drastic decrease in Recall (-12.1%). This confirms that direct LLM input on long-form documents suffers from context saturation, causing the model to skip clauses buried in the middle sections. Furthermore, disabling **Chain-of-Thought (CoT)** logic in the Extractor increases numerical hallucinations from 0.2% to 3.7%, highlighting the necessity of step-by-step reasoning when parsing intricate interest rate formulas.

TABLE II
ABLATION ANALYSIS OF FRAMEWORK COMPONENTS

Configuration	Recall	F1-Score	Halluc. Rate
Full Framework	**98.5%**	**98.1%**	**0.2%**
w/o Selector (Direct Full Text)	86.4%	89.2%	0.5%
w/o CoT Prompting	94.1%	91.2%	3.7%
w/o Symbolic Reasoner	95.8%	94.5%	1.8%

V. Qualitative Analysis and Case Studies

While quantitative metrics provide a high-level overview of performance, a granular qualitative analysis is essential to understand the framework's behavior in complex real-world scenarios.

A. Failure Modes and Error Analysis

Despite the high overall performance, we identified two primary failure modes:

- **Logical Parsing Failures**: In 0.8% of cases, the Extractor Agent failed to correctly parse the logical dependency between a base rate and a floating adjustment. For instance, in a corporate credit line agreement, the interest rate was defined as: "LPR + 50bps, provided that the net debt ratio remains below 2.0; otherwise, LPR + 150bps." Standalone LLMs often extracted a single value, ignoring the conditional switch. Our framework mitigated this by using the Compliance Reasoner to flag such ambiguous extractions for human review.
- **Boundary Delimitation Errors**: These occurred when the model included irrelevant preamble text within the extracted field. While this does not always affect the final compliance status, it increases the noise in the structured JSON output, particularly in documents with non-standard formatting or low OCR quality.

B. Framework Robustness and Success Cases

The *FinCompliance-LLM* framework demonstrates exceptional robustness in scenarios where traditional systems typically fail.

1) Cross-lingual Generalization: A major advantage of our approach is its zero-shot cross-lingual capability. The framework processed English SEC EDGAR filings and Chinese retail banking templates with the same underlying configuration. It successfully mapped diverse terms like "APR" and "Annualized Interest Rate" to the same internal symbolic representation, demonstrating that the agentic workflow effectively leverages the LLM's multi-lingual latent space.

2) Handling Complex Legal Inference: In a car loan agreement, the text stated "Prepayment is not allowed." While keyword-based systems (e.g., Regex) failed to find terms like "penalty" or "fee," the Extractor Agent correctly inferred that a prohibition on prepayment constitutes a structural restriction, allowing the Compliance Reasoner to flag the document for regulatory violation regarding consumer choice.

3) Neuro-symbolic Safety Net: A critical success case involved a contract where the LLM extracted an interest rate of "3.5%" when the actual text specified "3.5% per month." In a purely generative system, this would lead to a catastrophic compliance breach, as the annualized rate exceeds 42%. However, the **Compliance Reasoner** detected a mismatch between the extracted value and the required "annualized" metadata flag, triggering a validation error. This case proves that our symbolic logic layer acts as a vital safety net against generative hallucinations.

VI. Conclusion and Future Work

This paper demonstrates the effectiveness of Large Language Models in automating the compliance review of financial contracts. By achieving a state-of-the-art recall of 98.5% and a hallucination rate of 0.2% in zero-shot settings on real-world datasets, the proposed *FinCompliance-LLM* framework proves to be a robust tool for financial institutions. Our results highlight the model's superior ability to navigate the complex linguistic structures of Chinese legal documents and the intricate clauses of SEC EDGAR agreements. The implications of this work are significant for the RegTech sector, reducing the manual review workload and providing a robust solution for scalable and auditable applications [13].

While the current framework demonstrates significant potential, several avenues for investigation remain. A primary objective involves integrating multimodal Large Language Models (MLLMs) to process scanned PDF contracts, addressing challenges of complex visual layouts and non-linear structures. Furthermore, leveraging fine-tuned open-source backbones such as Qwen3-14B could mitigate data sovereignty and cost concerns for smaller financial entities. Future iterations will also expand the regulatory predicate library to encompass more intricate requirements, such as anti-money laundering (AML) protocols. Throughout these advancements, maintaining transparency and preserving a "human-in-the-loop" approach remains critical for high-stakes compliance adjudication.

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